# APARAJITO SAHA

rajsaha@umich.edu — LinkedIn — Website

## RESEARCH INTERESTS

Interactive perception, scene understanding, embodied search, instruction-following mobile manipulation

## EDUCATION

**Ph.D. Robotics**, University of Michigan, Ann Arbor — *Ann Arbor, MI, USA*
**Advisor:** Dr. Bernadette Bucher — *Aug 2026 — Exp. May 2030*

**M.S. Robotics**, University of Michigan, Ann Arbor — *Ann Arbor, MI, USA*
**Cumulative GPA:** 4.00/4.00 — *Aug 2024 — May 2026*

**B.S. Electrical & Computer Engineering + Computer Science**, Cornell University — *Ithaca, NY, USA*
**Cumulative GPA (double major):** 3.74/4.00 — *Aug 2018 — May 2022*

## RESEARCH EXPERIENCE

**Graduate Student Research Assistant**, Mapping & Motion Lab — *University of Michigan*
**Principal Investigator:** Dr. Bernadette Bucher — *Dec 2024 — Present*

- Formulated the *interactive object search* problem, in which a mobile manipulator must reason about necessary physical interactions with an unexplored indoor environment to eliminate occlusions, identify visual affordances and locate semantically-defined target objects
- Designed a robotic task benchmark for the interactive object search problem to evaluate geometric and semantic reasoning for high-level planning alongside fine-grained control capabilities for mobile manipulator systems (**P1**)
- Led implementation efforts for a photorealistic, physics-driven extension on Nvidia's IsaacSim platform that leverages procedural generation and 3D asset population for creating multi-room indoor scenes for interactive object search (**W1**)
- Advised a team of graduate and undergraduate students for deploying state-of-the-art perception and manipulation systems such as MASt3R-SLAM, RoboPoint and AnyGrasp on a Boston Dynamics Spot robot, with the objective of producing an open-source software stack for instruction-following mobile manipulation

**Undergraduate Research Assistant**, EmPRISE Lab — *Cornell University*
**Principal Investigator:** Dr. Tapomayukh Bhattacharjee — *Aug 2021 — May 2022*

- Engineered a prototype circuit board for managing real-time communication of tactile feedback from a whole-arm fabric skin sensor system, earning an acknowledgment for the contribution in B. Xu et. al. "CushSense: Soft, Stretchable, and Comfortable Tactile Sensing Skin for Physical Human-Robot Interaction" *International Conference on Robotics & Automation, 2024*

## PROFESSIONAL EXPERIENCE

**Visiting Scientist**, General Motors Research and Development — *Warren, MI, USA*
**Manager:** Dr. Guangze Li — *May 2026 — Sept 2026*

- Designed learning-based approaches for autonomous depackaging and warehouse inbound induction using bimanual manipulators

**Software Development Engineer**, Amazon Fulfillment Technologies & Robotics — *North Reading, MA, USA*
**Manager:** Todd Clemetson — *Jul 2022 — Aug 2024*

- Designed autonomous robotic movement solutions deployed to over 1 million mobile robotic drive units in Amazon's global warehouse network
- Led a zero-downtime serverless rearchitecture and migration effort for a critical production service responsible for computing and updating unavailable resources on warehouse floors
- Contributed to algorithms for storage and retrieval optimization used in Amazon's new Sequoia fulfillment centers, reducing delivery miss rates
- Developed a continuous improvement and deployment pipeline framework for the robotic movement team, leading to a 2 hour reduction in downtime and maintenance per service per week across 8 Tier-1 and Tier-2 services
- Participated in on-call rotations to resolve high-severity events, mentored interns and new engineers on engineering practices and projects, and held open office hours for operations managers to troubleshoot floor issues

## PUBLICATIONS

**Preprints**

**P1** **Aparajito Saha,** Zhen Hao Gan, Jinjia Guo, Jacob Skwirsk, Jeremy Acheampong, Anton Arapin, Chahyon Ku, Yue Hu, Nima Fazeli, Bernadette Bucher. *ForageBench: A Photorealistic, Physically Grounded Benchmark for Interactive Object Search.* In Submission. Project Website.

**Workshop Submissions**

**W1** **Aparajito Saha,** Zhen Hao Gan, Jinjia Guo, Jacob Skwirsk, Jeremy Acheampong, Anton Arapin, Chahyon Ku, Yue Hu, Nima Fazeli, Bernadette Bucher. *PhyGS: Physically-Grounded Controllable Scene Generation.* Workshop on Multi-Agent Embodied Intelligent Systems, **CVPR** 2026. Project Website.

**Technical Reports**

**TR1** **Aparajito Saha***, Amulya Khurana*, Krithik Ranjan*. *AroundSound: Digital Echolocation for the Visually Impaired.* CircuitCellar Issue 386, Sept. 2022. *(* denotes equal contribution)*

## TEACHING

**Graduate Student Instructor**, University of Michigan, Ann Arbor — *Ann Arbor, MI, USA*
**EECS 442:** Computer Vision *(240 students/ semester)* — *Aug 2025 — Dec 2025*

**Teaching Assistant**, Cornell University — *Ithaca, NY, USA*
**ECE 5725:** Design with Embedded Operating Systems *(80 students/ semester)* — *Aug 2021 — May 2022*

**Peer Advisor**, Cornell University — *Ithaca, NY, USA*
**ENGRG 1050:** Engineering Seminar *(25 students/ semester)* — *Aug 2021 — May 2022*

## HONORS & GRANTS

**Robotics Department Fellowship**, University of Michigan, Ann Arbor — *2026 — 2027*
**U-M Library Student Mini-Grant**, University of Michigan, Ann Arbor — 2026
**Best Poster Presentation**, University of Michigan Robotics Symposium — *2026*
**Bachelor of Science, Cum Laude**, Cornell University — *2022*
**Departmental Honors in Computer Science**, Cornell University — *2022*
**Dean's List (Engineering)**, Cornell University — *2018 — 2022*

## ACADEMIC SERVICE

**Peer Reviewer** - Conference on Robot Learning (CoRL), International Conference on Robotics & Automation (ICRA)

## MENTORSHIP

**University of Michigan, Ann Arbor**

- Jinjia Guo, M.S. Robotics — *Sept 2025 — Dec 2025*
- Kornvik Tanpipat, M.S. Robotics — *Sept 2025 — Dec 2025*
- Saumit Vedula, M.S. Robotics — *Sept 2025 — Dec 2025*
- Aileen Yan, M.S. Electrical & Computer Engineering — *Sept 2025 — Dec 2025*
- Chenrui Gao, B.S. Computer Science & Engineering — *Sept 2025 — Dec 2025*
- Georgia Zender, B.S. Computer Science & Engineering — *Feb 2025 — May 2025*

**Amazon Fulfillment Technologies & Robotics**

- Hemant Palavali, Software Development Engineer — *December 2023 — March 2024*

**Cornell University**

- Cornell Cup Robotics Team (20+ students) — *Aug 2021 — May 2022*

## SKILLS

**Programming Languages:** Python, C++, Java, MATLAB
**Software:** IsaacLab, PyTorch, ROS2, Slurm, Git, Docker, AWS Lambda, AWS DynamoDB
**Hardware:** NVIDIA Jetson, Raspberry Pi, Rapid Prototyping (3D Printing, Soldering, Circuit Board Design)